

Build a Chatbot Using Hugging Face Transformers

```
import torch

from transformers import AutoTokenizer, AutoModelForCausalLM

model_name = "Qwen/Qwen2.5-1.5B-Instruct"

print("Loading model... ⌚ ")

tokenizer = AutoTokenizer.from_pretrained(model_name)

model = AutoModelForCausalLM.from_pretrained(
    model_name,
    torch_dtype=torch.float32
)

model.to("cpu")

model.eval()

print("✅ Model loaded successfully!")

messages = [
    {
        "role": "system",
        "content": "You are a helpful AI assistant."
    }
]

def chatbot(user_input):
    global messages

    messages.append({"role": "user", "content": user_input})

    text = tokenizer.apply_chat_template(
        messages,
        tokenize=False,
```

```

        add_generation_prompt=True
    )
    inputs = tokenizer(text, return_tensors="pt")
    with torch.no_grad():
        outputs = model.generate(
            **inputs,
            max_new_tokens=100,
            temperature=0.7,
            top_p=0.9,
            do_sample=True,
            pad_token_id=tokenizer.eos_token_id
        )
    response = tokenizer.decode(
        outputs[0][inputs["input_ids"].shape[-1]:],
        skip_special_tokens=True
    ).strip()
    messages.append({"role": "assistant", "content": response})
    return response

print("\n 🤖 Chatbot Ready (CPU)")
print("Type 'exit' to quit | 'clear' to reset\n")
while True:
    user_input = input("You: ")
    if user_input.lower() in ["exit", "quit", "bye"]:
        print("Bot: Goodbye!")
        break
    if user_input.lower() == "clear":

```

```

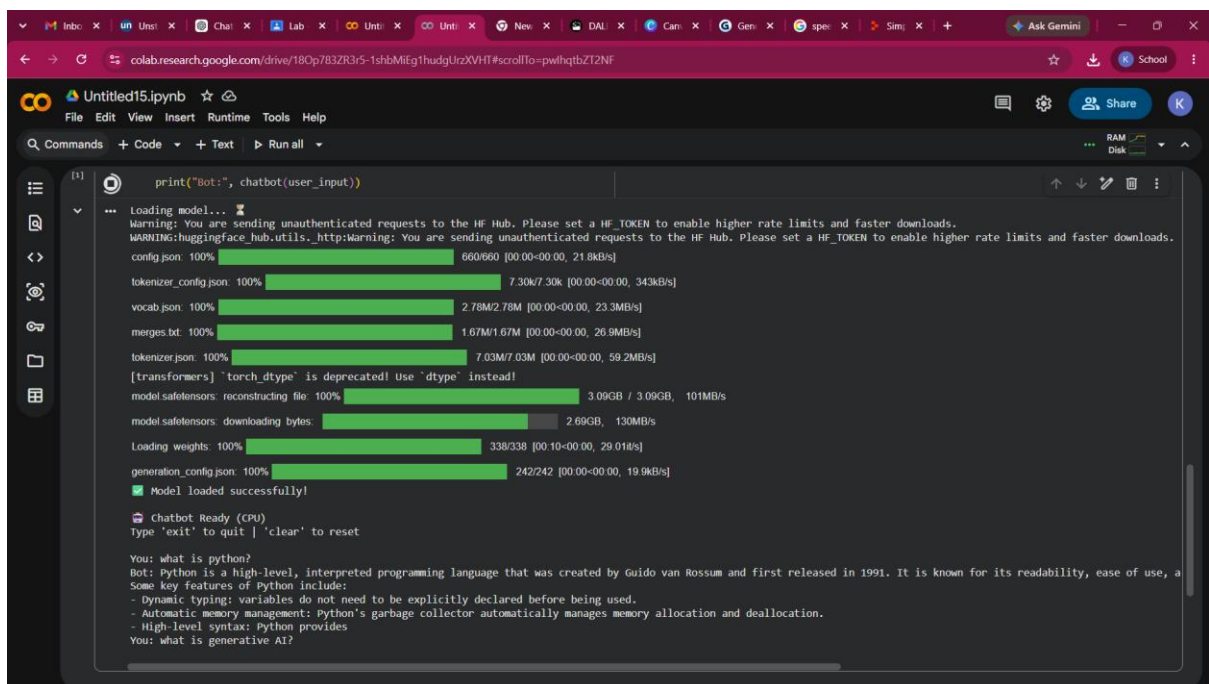
messages = [{"role": "system", "content": "You are a helpful AI assistant."}]

print("Bot: Cleared.")

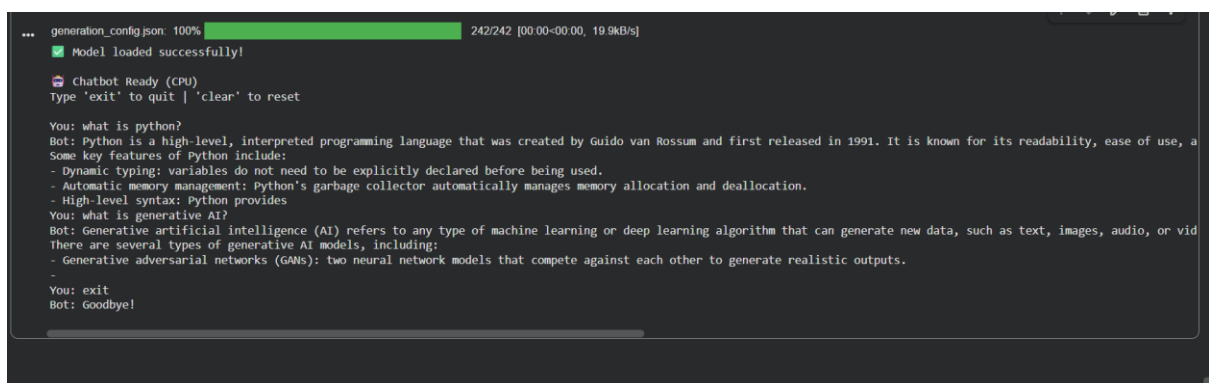
continue

print("Bot:", chatbot(user_input))

```



The screenshot shows a Google Colab notebook titled 'Untitled15.ipynb'. The code cell contains a single line: `print("Bot:", chatbot(user_input))`. The output shows the progress of loading a model, with progress bars for various components like config.json, tokenizer_config.json, vocab.json, merges.txt, tokenizer.json, model.safetensors, and generation_config.json. A warning message from HuggingFace is also visible. The chatbot interface is ready, displaying the prompt 'You: what is python?' and the bot's response: 'Bot: Python is a high-level, interpreted programming language that was created by Guido van Rossum and first released in 1991. It is known for its readability, ease of use, a Some key features of Python include: - Dynamic typing: variables do not need to be explicitly declared before being used. - Automatic memory management: Python's garbage collector automatically manages memory allocation and deallocation. - High-level syntax: Python provides



The screenshot continues the output from the previous image. The chatbot responds to 'You: what is generative AI?' with: 'Bot: Generative artificial intelligence (AI) refers to any type of machine learning or deep learning algorithm that can generate new data, such as text, images, audio, or vid There are several types of generative AI models, including: - Generative adversarial networks (GANs): two neural network models that compete against each other to generate realistic outputs. -'. The user then enters 'You: exit' and the bot responds 'Bot: Goodbye!'